Tutorial 7: Hypothesis testing (cont.)

1. In a study about the effect of using a mobile phone while driving, the reaction times of $n_1 = 31$ drivers using a phone were measured and gave a mean response time of 633.5 milliseconds (with a standard deviation of 98.1 milliseconds). The reaction times of a second group of $n_2 = 31$ drivers not using a phone (the *control* group) were also measured, and gave a mean response time of 571.2 milliseconds (with a standard deviation of 89.8 milliseconds). All drivers were chosen and allocated to the groups randomly.

   (a) By calculating a suitable test statistic and determining the relevant critical region, test the hypothesis that using a mobile phone while driving increases the mean reaction time (at $\alpha = 0.01$).

   (*Hint: You first need to specify H_0 and H_1. Think whether the hypothesis to be tested would be more conveniently set as H_0 or H_1.*)

   (b) Calculate the p-value for the test in (a), give its interpretation and state your conclusion regarding the hypothesis for a 0.01 significance level.

2. Two random samples are selected from two normal populations. For the first sample we have $n_1 = 9, \sum_i x_i = 180$ and $\sum_i x_i^2 = 4000$, while for the second sample $n_2 = 10, \sum_i y_i = 220$ and $\sum_i y_i^2 = 4930$. Test the hypothesis $H_0 : \sigma_x^2 = \sigma_y^2$ against $H_1 : \sigma_x^2 > \sigma_y^2$ (at $\alpha = 0.05$).

3. Consider the data in Question 4 of Tutorial 5 on the coefficient of thermal expansion of a polymer.

Method A	Method B
6.8	7.2
8.4	8.3
7.9	8.3
6.5	7.5
9.5	9.5
7.0	7.7
9.1	9.4
7.5	7.8
9.8	10.3

   (a) Assume that the samples used in the two methods are independent, and test the hypothesis that there is no difference between the mean for method B and the mean for method A at the 0.05 level of significance.

   (b) Test the same hypothesis when the data in the two samples are matched pairs.

   (c) How do your conclusions here relate to the confidence intervals obtained in Tutorial 5?

4. For a certain class of policies insured by a large insurance company it is believed that the probability of each policy to give rise to any claims is 0.5, independently of all other policies. A random sample of 150 such policies is selected and 83 are found to have given rise to claims.

   By calculating an appropriate p-value, test the hypothesis (at $\alpha = 0.01$) that the probability of a policy to give rise to claims is greater than 0.5.

   (*Hint: use the normal approximation to the binomial distribution.*)

5. In November 2019 a random sample of 3000 consumers were asked whether they would prefer a specific brand of a food product over other brands, and 7.7% of them answered that they would. In a similar survey in November 2020, 5.9% of the consumers in a sample of 3120 (from the same geographical population) indicated that they would prefer this brand. Use a suitable test to investigate (at $\alpha = 0.05$) if there is a change in the preferences of consumers between November 2019 and November 2020.

Exam Style Question

One way to measure a person's fitness is to measure their body fat percentage. Data are measured for a random selection of people who did two different workout plans at a gym three times a week for a three-month period; at the end of which their trainer measured the body fat. The table below shows the data.

Plan A	8.5	20.8	23.1	22.6	24.1	15.5	15.4	13.7
Plan B	24.6	6.8	10.4	10.3	16.4	8.3	13.2	17.7

(a) Perform a hypothesis test to check whether or not there is a difference between the mean body fat percentage for Plan A and the mean body fat percentage for Plan B. Make sure you state the p-value and give a clear conclusion. You are told that $s_p^2 = 32.51$.

[5 marks]

(b) Perform a hypothesis test to check whether or not the population variances are equal with a significance level of 95%. You are told that $s_A^2/s_B^2 = 0.89$.

(3 marks)

(c) What are the main assumptions in assessing whether or not the test of part (a) is valid? Give reasons whether or not these assumptions hold for the above data.

[2 marks]

[10 marks]